

Assessment of mitochondrial respiratory chain enzymatic activities on tissues and cultured cells

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The assessment of mitochondrial respiratory chain (RC) enzymatic activities is essential for investigating mitochondrial function in several situations, including mitochondrial disorders, diabetes, cancer, aging and neurodegeneration, as well as for many toxicological assays. Muscle is the most commonly analyzed tissue because of its high metabolic rates and accessibility, although other tissues and cultured cell lines can be used. We describe a step-by-step protocol for a simple and reliable assessment of the RC enzymatic function (complexes I–IV) for minute quantities of muscle, cultured cells and isolated mitochondria from a variety of species and tissues, by using a single-wavelength spectrophotometer. An efficient tissue disruption and the choice for each assay of specific buffers, substrates, adjuvants and detergents in a narrow concentration range allow maximal sensitivity, specificity and linearity of the kinetics. This protocol can be completed in 3 h.

INTRODUCTION

Mitochondria perform crucial cellular reactions, including the production of energy through the mitochondrial RC, the regulation of cell death, calcium metabolism and the production of reactive oxygen species. The RC comprises four enzymatic complexes (complexes I–IV) embedded in the inner mitochondrial membrane, which catalyze the transfer of reducing equivalents from high-energy compounds produced by the reactions of the Krebs cycle to oxygen, with the ultimate production of an electrochemical gradient through the inner mitochondrial membranes to drive the synthesis of ATP by ATP synthase.

Dysfunction of the mitochondrial RC is a key player in a variety of human disorders, including primary mitochondrial diseases caused by mutations both in the mitochondrial and nuclear DNA¹, as well as in common conditions, such as aging², diabetes³, cancer^{4,5}, drug toxicity^{6,7}, and several neurodegenerative diseases⁸, including Parkinson's disease⁹ and Alzheimer's disease¹⁰.

The enzymatic activities of RC complexes I–IV are assayed spectrophotometrically, and the results are commonly normalized to the total muscle protein content or to the activity of citrate synthase (CS)¹¹, a mitochondrial matrix enzyme. Although a potential limitation of these assays is that the enzymatic activities are measured using *in vitro* conditions that are not physiological in terms of pH, osmolarity, substrate concentrations and cellular context and do not allow for the evaluation of respiratory coupling, they still provide crucial quantitative information concerning the maximal catalytic activities of the RC complexes; they are also easy to reproduce and can be performed using frozen tissue and cellular samples.

An ideal investigation of the mitochondrial respiratory function should also include a combination of other assays (such as polarographical measurements of oxygen consumption in intact isolated mitochondria¹² or permeabilized cells or tissues¹³, ATP synthesis¹⁴, oxidation studies with radiolabeled substrates¹⁵ and the determination of mitochondrial membrane potential¹⁶ that may provide a sensitive detection of mitochondrial dysfunction. However, these techniques have the disadvantage of requiring fresh samples with intact mitochondrial membranes; they are more complex and time consuming, and they require specific equipment that is not

commonly found in all laboratories. Therefore, spectrophotometric assays remain a first-line technique both for research studies on mitochondrial disorders and for diagnostics.

In recent years, it has become apparent that many published protocols for spectrophotometric assays were suboptimal, because complex biochemical interferences led to enzymatic inhibition or to insufficient linearity of the kinetics with time¹⁷ or with respect to protein concentrations¹⁸. These analytical pitfalls can severely impair sensitivity and precision, leading to analytical inconsistencies¹⁹ and remarkable variation of results between laboratories. Our protocols were developed to overcome or limit most of these problems, as well as to improve the sensitivity, specificity and precision while maintaining procedural simplicity¹⁷. The protocols described herein detail the procedure for mitochondrial RC enzyme activity analysis using minute quantities of mammalian muscle tissue and cultured skin fibroblasts, using substantially smaller amounts of samples compared with previously published protocols²⁰. In our experience, these methods can also be applied to other tissues, such as heart, liver and brain, as well as to several cell lines (such as HeLa), cultured myotubes²¹, platelets²² and isolated mitochondria from other organisms, such as yeast and *Caenorhabditis elegans*. If tissue homogenates are used, the sample preparation phase must be optimized for each tissue. For example, hard tissues such as skeletal muscle require a harsher homogenization (i.e., with the use of a glass-glass tissue grinder) than softer tissues, such as brain or liver (i.e., with use of a Teflon-glass tissue grinder). An efficient lysis procedure should lead to an efficient disruption of cellular membranes (both plasma membranes and mitochondrial membranes) in order to solubilize the mitochondrial enzymes and make them accessible to the specific substrates without chemical or physical inactivation of RC enzymes, which are sensitive to chemico-physical treatments. For example, cytochrome *c* oxidase is exquisitely sensitive to mechanical forces during cell lysis and is easily damaged by sonication²³ or extensive homogenization¹⁷, whereas the activity of complexes I and II is maximized under the same conditions.

Although the use of isolated mitochondria is mandatory for the analysis of complex I activity in cultured cells, isolation of

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mitochondria requires a larger quantity of fresh tissue or cultured cells^{20,24}, which might not be available, especially in human studies. Moreover, incomplete recovery of mitochondria during isolation has the potential disadvantage of selecting different mitochondrial populations (i.e., subsarcolemmal versus intermyofibrillar mitochondria in muscle), which might lead to a potential bias in the results²⁵. Therefore, our approach aimed to develop sensitive and reliable assays for assessing RC enzymes in tissue and cell homogenates to reduce the sample quantity required for the analysis while maintaining high precision and specificity.

Some additional advantages are derived from the use of sample homogenates rather than isolated mitochondria for the assessment of RC function, including (i) the reduction of time and costs; (ii) the use of frozen tissue samples, which are not suitable for isolation of mitochondria²³; and (iii) the possibility of estimating mitochondrial abundance in the examined tissue/cell line by calculating the activity of the mitochondrial matrix enzyme CS normalized to total tissue protein content.

We did not include a spectrophotometric method for complex V (ATP hydrolysis) because of its insufficient reliability in frozen muscle²⁶ and in cultured cells, which is caused by its high oligomycin-resistant activities²⁷. ATPase activity can be measured indirectly by polarographic assays or blue-native gel electrophoresis followed by in-gel assay of ATPase, and ATP synthesis can be estimated using the luciferase-luciferin assay in permeabilized cells with a luminometer¹⁴.

Experimental design

Mitochondrial RC enzymatic activities can be measured in a variety of mammalian tissues (i.e., skeletal muscle, heart, brain, liver) and cells, both in crude homogenates and in isolated mitochondria. Moreover, these protocols can also be applied to other organisms, such as yeast or *C. elegans*, using isolated mitochondria prepared as previously described^{28,29}. Step 1A describes the preparation of mouse skeletal muscle homogenates. The same protocol can also be applied successfully to human and bovine muscles. All assays can be reliably performed on muscle homogenates when prepared as indicated, without requiring the isolation of mitochondria.

Freezing of the sample is included for convenience and for disrupting the mitochondrial membranes in order to make the substrates accessible to the enzymes, which represents a crucial step for maximizing the enzymatic activities of complexes I and II. However, a partial loss of activity might be observed for complex II + III after freezing the muscle. Therefore, it is important to treat all the samples similarly before the analysis. Nevertheless, mitochondria can be used after isolation from fresh muscle and from a variety of other tissues according to Frezza *et al.*²⁴.

For the assessment of RC enzymes in cultured cells, all assays except those for complexes I and I + III, can be performed on cell lysates prepared as detailed in Step 1B. However, for the analysis of complex I activity in cultured cells, it is imperative to use enriched mitochondrial fractions in order to reduce the overwhelming nonspecific

TABLE 1 | Conditions for spectrophotometric assays of respiratory chain enzymes and citrate synthase activities in tissues and cells.

	CI	CII	CIII	CIV	CI + III	CII + III	CS
Option in Step 2	A	B	C	D	E	F	G
λ (nm)	340	600	550	550	550	550	412
ϵ (mmol ⁻¹ cm ⁻¹)	6.2	19.1	18.5	18.5	18.5	18.5	13.6
Buffer	KP, 50 mM	KP, 25 mM	KP, 25 mM	KP, 50 mM [25 mM]	KP, 50 mM	KP, 20 mM [100 mM]	Tris, 100 mM
pH	7.50	7.50	7.50	7.00	7.50	7.50	8.00
Substrates/ electron acceptors	NADH, 100 μ M Ub ₁ , 60 μ M	Succinate, 20 mM DCPIP, 80 μ M DUB, 50 μ M	DubH ₂ , 100 μ M Cyt c, 75 μ M	Cyt c H ₂ , 60 μ M [50 μ M]	NADH, 200 μ M Cyt c, 50 μ M	Succinate, 10 mM Cyt c, 50 μ M	DTNB, 100 μ M Ac CoA, 300 μ M
Detergent	—	—	Tween-20 (0.025% (vol/vol))	—	—	—	Triton X-100 (0.1% (vol/vol))
Other reagent(s)	BSA, 3 mg ml ⁻¹ KCN, 300 μ M	BSA, 1 mg ml ⁻¹ KCN, 300 μ M	KCN, 500 μ M EDTA, 100 μ M	—	BSA, 1 mg ml ⁻¹ KCN, 300 μ M	KCN, 300 μ M	—
Specific Inhibitor	Rotenone 10 μ M	Malonate, 10 mM, or TTFA, 500 μ M	Antimycin A, 10 μ g ml ⁻¹	KCN, 300 μ M	Rotenone, 10 μ M	Malonate, 10 mM	—

Abbreviations: λ , selected wavelength for the assay; ϵ , extinction coefficient; Ac CoA, acetyl coenzyme A; BSA, fatty acid-free bovine serum albumin; Cyt c, cytochrome c; Cyt c H₂, reduced cytochrome c; DCPIP, 2,6-dichlorophenolindophenol; DUB, decylubiquinol; DubH₂, decylubiquinol; DTNB, 5,5'-dithiobis(2-nitrobenzoic acid); KCN, potassium cyanide; KP, potassium phosphate buffer; Tris, Tris buffer; TTFA, 2-thenoyltrifluoroacetone; Ub₁, ubiquinone₁.

Concentrations in square brackets, where present, indicate the application for cultured cell lines.

Bold square brackets, where present, indicate minor modifications applied to complex IV and complex II+III assays in cells to increase the sensitivity.

rotenone-insensitive activities. In our experience, complex I+III cannot be reliably assayed in mitochondria from cultured cells and from liver, because of high rotenone-resistant activities. The method for mitochondrial isolation from cultured cells, described by Janssen *et al.*³⁰, is summarized with only minor modifications in Step 1C.

All assays are performed with a single-wavelength, temperature-controlled spectrophotometer at 37 °C and should be run in duplicate or triplicate. Ideally, an aliquot of the same control

sample should be included as an internal control in experiments run on different days, along with the samples to be analyzed. The biochemical conditions of the RC assays for solid tissues and cells are summarized in **Table 1**. The protocols for these two applications are similar. However, some minor modifications have been applied to complex IV and complex II + III assays in cells to increase the sensitivity, as highlighted (in bold in squared brackets) in **Table 1**.

MATERIALS

REAGENTS

- Trypsin-EDTA solution (0.05% (wt/vol); Invitrogen, cat. no. 253000-054)
- Penicillin (10,000 U ml⁻¹)/streptomycin (10,000 µg ml⁻¹) (Invitrogen, cat. no. 15140-122)
- 2-Thenoyltrifluoroacetone (TTFA; Sigma, cat. no. T27006)
- 2,6-Dichlorophenolindophenol sodium salt hydrate (DCPIP; Sigma, cat. no. 33125)
- L-Glutamine (200 mM; Invitrogen, cat. no. 25030-024)
- 5,5'-Dithiobis(2-nitrobenzoic acid) (DTNB; Sigma, cat. no. D218200)
- Acetyl CoA lithium salt (Ac CoA; Sigma, cat. no. A2181) **▲ CRITICAL** Store at -20 °C.
- Antimycin A (Sigma, cat. no. A8674) **▲ CRITICAL** Store at -20 °C.
- Ascorbic acid (Sigma, cat. no. A5960)
- β-Nicotinamide adenine dinucleotide, reduced dipotassium salt (NADH; Sigma, cat. no. N4505) **▲ CRITICAL** Keep the powder tightly closed and dry in a desiccator at 4 °C. Oxidation of NADH will result in a color shift from light yellow-white to dark yellow.
- Bovine serum albumin, essentially fatty acid free (Sigma, cat. no. A6003) **▲ CRITICAL** Store at 4 °C.
- Cytochrome *c* from equine heart (Sigma, cat. no. C2506). Bovine cytochrome *c* can be used as an alternative without major differences **▲ CRITICAL** Store at -20 °C. **▲ CRITICAL** The use of cytochrome *c* extracted with trichloroacetic acid will result in increased rates of complex III (both specific and aspecific) compared with those obtained with the use of cytochrome *c* extracted with acetic acid³¹ (Sigma, cat. no. C7752). Always use the same type of cytochrome *c* for consistent results for complex III assay.
- Decylubiquinone (DUB; Sigma, cat. no. D7911) **▲ CRITICAL** Store at -20 °C.
- Dulbecco's modified Eagle's medium (Invitrogen, cat. no. 419656)
- EDTA (Sigma, cat. no. E1644)
- Ethylene glycol-bis(2-aminoethylether)-N,N,N',N'-tetraacetic acid (EGTA; Sigma, cat. no. E4378)
- Fetal bovine serum (Invitrogen, cat. no. 10270-106)
- Malonic acid (Sigma, cat. no. M1296)
- Muscle tissue, stored at -80 °C, or cultured fibroblasts (see REAGENT SETUP)
- Oxaloacetic acid (Sigma, cat. no. O4126) **▲ CRITICAL** Store at -20 °C.
- Phosphate-buffered saline without Ca²⁺ and Mg²⁺ (PBS; Invitrogen, cat. no. 14200-067)
- Potassium borohydride (Sigma, cat. no. 438472) **▲ CRITICAL** Store in a desiccator.
- Potassium chloride (KCl; Sigma, cat. no. P9333)
- Potassium cyanide (KCN; Sigma, cat. no. 60178)
- Potassium ferricyanide (Sigma, cat. no. 702587)
- Potassium phosphate dibasic (Sigma, cat. no. P2222)
- Potassium phosphate monobasic (Sigma, cat. no. P5655)
- Rotenone (Sigma, cat. no. R8875) **▲ CRITICAL** Keep protected from light.
- Sodium hydrosulfite (Sigma, cat. no. 157953) **▲ CRITICAL** Keep the container tightly closed and dry.
- Succinic acid (Sigma, cat. no. S7501)
- Sucrose (Sigma, cat. no. 84100)
- Tris(hydroxymethyl)aminomethane (Tris; Sigma, cat. no. 154563)
- Triton X-100 (Sigma, cat. no. 157953)
- Tween-20 (Sigma, cat. no. P7949)
- Ubiquinone₁ (Sigma, cat. no. C7956) **▲ CRITICAL** Store at -20 °C.
- Distilled water

- HCl
- KOH
- Ethanol
- Alconox (Sigma; cat. no. Z273228)

EQUIPMENT

- Dishes for cell culture (150 cm²)
- Scalpel with removable blades
- Tightly fitting glass/glass tapered conical tissue grinder (1 ml; Wheaton Science products, cat. no. 358103; **Fig. 1a–c**) **▲ CRITICAL** Verify carefully that there is a tight and homogeneous compliance between the glass pestle and mortar, and that the tip of the pestle touches the end of the mortar (**Fig. 1a,b**). Even tiny differences in the shape of the individual glass components will require a careful matching between mortars and pestles. Defects or variability in the compliance of the tissue grinders (**Fig. 1c**) are not uncommon; they may cause variations in the shearing forces and in the efficacy of the homogenization process, leading to inconsistent results.
- Tightly fitting glass/Teflon tissue grinder (2 ml; Kartell Labware Division, cat. no. 6102; **Fig. 1d,e**) **▲ CRITICAL** The same concerns regarding the compliance of the homogenizers also apply to glass-Teflon tissue grinders produced by some manufacturers. The tip of the pestle should touch the end of the mortar (**Fig. 1d**). Products showing a poor compliance should not be used (**Fig. 1e**).
- Stirrer motor with electronic speed controller (Cole-Parmer, cat. no. EW-04369-25)
- Polypropylene tubes (50 ml)
- Polypropylene tubes (14 ml)
- Microcentrifuge tubes (1.5 ml)
- Refrigerated centrifuge for microcentrifuge tubes
- PD10 disposable desalting column (GE Healthcare, cat. no. 17-0851-01)
- Hamilton syringe (50 µl; Hamilton, cat. no. 705 N)
- Ultraviolet-visible spectrophotometer with temperature control (Beckman-Coulter, cat. no. DU 800 or Varian Cary 100 Bio UV-Vis)
- Disposable 1-ml ultraviolet-visible cuvettes with 1-cm path length
- Container for liquid nitrogen
- Cell counting chamber
- Parafilm

REAGENT SETUP

Fibroblast culture medium Use Dulbecco's modified Eagle's medium supplemented with 10% (vol/vol) fetal bovine serum, 2 mM L-glutamine and penicillin/streptomycin (100 U ml⁻¹ penicillin/100 µg ml⁻¹ streptomycin). Store at 4 °C for up to 10 d.

Human cultured fibroblasts Two or three days before performing the experiments, plate the fibroblasts on 150-cm² dishes so as to have the desired number of cells on the day planned for the analysis (5 × 10⁶ fibroblasts for RC analysis on total cell lysates, 10–15 × 10⁶ for isolated mitochondria).

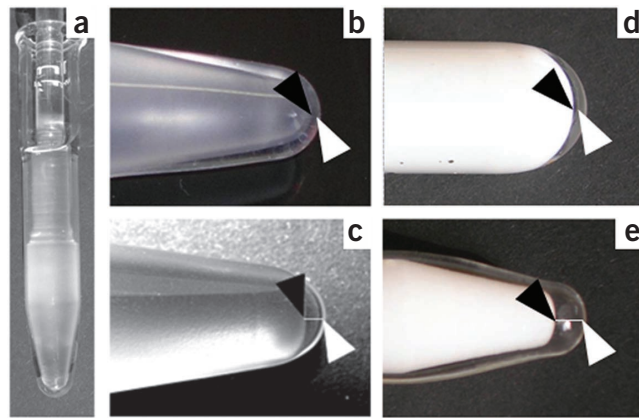
▲ CRITICAL Fibroblasts should be 80–90% confluent. **! CAUTION** Cell lines can be purchased from ATCC (<http://www.lgcstandards-atcc.org/>). If skin or muscle biopsies are performed on human subjects, informed consent must be obtained before the procedure and all relevant ethical regulations should be adhered to.

Mouse skeletal muscle Skeletal muscle should be immediately removed after euthanizing the mouse by cervical dislocation. It should be dissected in small fragments of about 30–50 mg, frozen in liquid nitrogen and stored at -80 °C.

▲ CRITICAL Institutional and governmental regulations on animal care and handling vary. Ensure that you hold the appropriate authorization to conduct animal experiments.

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Figure 1 | Potter tissue grinders for tissue and cell homogenization. (a) Glass-glass conical tissue grinder for muscle homogenization. The conical shape of the grinder improves the yield of the muscle homogenization. (b) Good compliance between mortar and pestle. There is no gap between the tip of the pestle (black arrowhead) and the internal end of the mortar (white arrowhead). (c) Poor compliance between pestle and mortar. There is a gap between the tip of the pestle and the bottom of the vial (indicated by the white line), which can lead to potential inconsistency in the homogenization efficacy, especially when homogenizing small muscle samples. (d) Good compliance between the Teflon pestle and the glass vial in a glass-Teflon tissue grinder, useful for homogenization of cultured cells, liver and brain tissue. (e) Defective compliance between the Teflon pestle and the glass vial.



Sucrose muscle homogenization buffer (250 mM) Prepare 50 ml of muscle homogenization buffer without sucrose by dissolving 0.121 g of Tris, 0.15 g of KCl and 0.038 g of EGTA in distilled water. Adjust the pH to 7.4 and the volume to 50 ml. Store at 4 °C for up to 2 months. Add 0.854 g of sucrose to 10 ml of this buffer on the same day of the experiment and vortex thoroughly until dissolved. Keep the sucrose muscle homogenization buffer on ice.

Potassium phosphate buffer (0.5 M, pH 7.5) Titrate 0.5 M potassium phosphate dibasic with 0.5 M potassium phosphate monobasic up to a pH of 7.5. Store at 4 °C for up to 2 months.

Potassium phosphate buffer (0.1 M, pH 7.0) Titrate 0.1 M potassium phosphate dibasic with 0.1 M potassium phosphate monobasic up to a pH of 7.0. Store at 4 °C for up to 2 months.

Tris (200 mM, pH 8.0) with Triton X-100 (0.2% (vol/vol)) Dissolve 1.21 g of Tris in 40 ml of distilled water, adjust the pH to 8.0 with HCl, add 0.1 ml of Triton X-100 and adjust the volume to 50 ml. Store at 4 °C up to 2 months.

Box 1 | Preparation of a 1 mM reduced cytochrome c solution

This can be prepared in two ways. Method 1 allows for the solution to be prepared in advance. Method 2 uses an extemporaneous reduction of a 1 mM cytochrome c solution with sodium dithionite, and if this method is used it should be prepared freshly as needed.

Method 1

1. Dissolve 110 mg of ascorbic acid in 1 ml of 10 mM potassium phosphate buffer (pH 7.0), and then add a few grains of Tris powder to obtain a pH of 6.5–6.8.

▲ **CRITICAL** The pH of the ascorbic acid solution should not be higher than 6.8.

2. Dissolve 250 mg of cytochrome c in 1.2 ml of 10 mM potassium phosphate buffer (pH 7.0), and then add 0.3 ml of ascorbic acid solution.

3. Incubate with agitation for 1 h at 4 °C. The reduction of cytochrome c will result in a color shift from brown to pink-orange.

4. Purify from excessive ascorbic acid in solution using a PD10 disposable desalting column previously equilibrated with 50 ml of 10 mM potassium phosphate buffer (pH 7.0). Discard the first three drops, and then collect the eluted solution of reduced cytochrome c in several 1.5-ml microtubes (about seven drops in each vial). Store the vials on ice.

5. To check the efficiency of cytochrome c reduction and to assess the absence of residual ascorbic acid in the solution, read for 2 min the absorbance of a 50 µM cytochrome c solution in a 1-ml cuvette at 550 nm. The decrease in absorbance should not exceed 0.005 per min.

6. Add to the cuvette 1 µl of a diluted potassium ferricyanide solution obtained by dissolving a few crystals of ferricyanide in 10 ml of water under a hood: this will result in cytochrome c oxidation (Fig. 2, white arrowheads). Read for 3 min.

! **CAUTION** Potassium ferricyanide is highly toxic; avoid skin contact and inhalation.

▲ **CRITICAL** The absorbance values should be stable (Fig. 2a). Any increment indicates the presence of residual ascorbic acid in solution (Fig. 2b), whereas a decrement in absorbance greater than 0.005 OD units per minute indicates excessive auto-oxidation.

7. Add a few crystals of sodium dithionite (Fig. 2, black arrowheads) and read again: the initial absorbance values before the addition of dithionite should be about 95% of the values after dithionite addition. Discard any vial showing signs of autoreduction and pool together the selected vials. Aliquots of this solution can be stored under liquid nitrogen for several years.

8. Before use, dilute with distilled water an aliquot of reduced cytochrome c to a concentration of 1 mM. Effective cytochrome c reduction should be checked by calculating the ratio of the absorbance values at 550 nm versus 565 nm of this solution at a final concentration of 20 µM. A ratio greater than 6 will indicate that the cytochrome c has not reoxidized and that it is ready to be used³⁵.

Method 2

1. Dissolve 12.5 mg of oxidized cytochrome c in 1 ml of 20 mM potassium phosphate buffer (pH 7.0).

2. Reduce the cytochrome c solution with a few grains of sodium dithionite on the tip of a pipette, just before use. The solution will change color from brown (Fig. 3a) to orange-pink (Fig. 3b). Vortex thoroughly.

3. Effective cytochrome c reduction should be checked by calculating the ratio of the absorbance values at 550 nm versus 565 nm of this solution at a final concentration of 20 µM. A ratio greater than 6 will indicate effective cytochrome c reduction.

▲ **CRITICAL** If this method is being used, the solution should be freshly prepared.

▲ **CRITICAL** The amount of sodium dithionite should be the least sufficient to achieve the color shift; however, avoid excessive dithionite in solution in order to avoid potential inhibition of cytochrome c oxidase activity.

See the TROUBLESHOOTING section for more information.

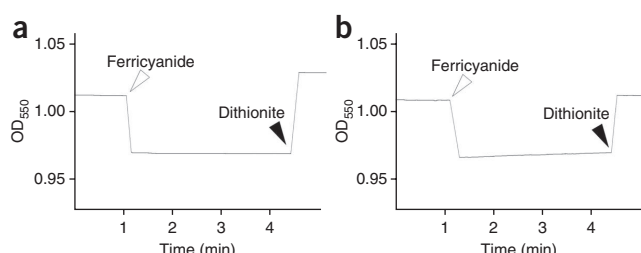


Figure 2 | Quality control of 50 μM reduced cytochrome *c* solution. After adjusting the concentration of cytochrome *c* to reach an optical density at 550 nm (OD_{550}) of ~ 1 , addition of 1 μl of a very diluted potassium ferricyanide solution (white arrowheads) results in partial oxidation of the reduced cytochrome *c* solution. Absorbance values should be followed for 2–3 min, and should not further decrease more than $0.005/\text{OD U min}^{-1}$. Any increments in OD are indicative of the presence of reducing substances (ascorbic acid) in the solution, precluding the use of that particular fraction. Addition of sodium dithionite (black arrowheads) completely reduces cytochrome *c*. The final absorbance should not be greater than 105% of the initial value. **(a)** Good-quality cytochrome *c* preparation. **(b)** The presence of residual ascorbate in the preparation causes reduction of cytochrome *c* after addition of ferricyanide.

Ubiquinone, solution (10 mM) Dissolve 2 mg of ubiquinone₁ in 0.8 ml of absolute ethanol and store it in 100- μl aliquots at -20°C for several months.

BSA (50 mg ml⁻¹) Dissolve 250 mg of defatted BSA in 5 ml of distilled water and store it in 1-ml aliquots at 4°C for up to 1 month or until signs of microbial contamination are seen.

KCN solution (10 mM) Dissolve 6.5 mg of KCN in 10 ml of distilled water under a fume hood. **▲ CRITICAL** This should be freshly prepared.

! CAUTION KCN is highly toxic; avoid skin contact and inhalation.

NADH solution (10 mM) Dissolve 7.5 mg of NADH in 1 ml of distilled water. **▲ CRITICAL** This should be freshly prepared.

Rotenone solution (1 mM) Dissolve 3.94 mg of rotenone in 10 ml of absolute ethanol and store it in 1-ml aliquots protected from light at -20°C for several months. **! CAUTION** Rotenone is highly toxic; avoid skin contact and inhalation.

Reduced cytochrome *c* solution (1 mM) See Box 1 and Figures 2 and 3 for details on how to prepare this solution.

Oxidized cytochrome *c* solution (1 mM) Dissolve 12.5 mg of cytochrome *c* in 1 ml of distilled water (Fig. 3a).

Succinic acid solution (400 mM) Dissolve 2.36 g of succinic acid in 20 ml of distilled water and adjust the pH to 7.4 with 3 M KOH; next, adjust the volume to 50 ml with distilled water. Store in 1-ml aliquots at -20°C for several months.

DCPIP solution (0.015% (wt/vol)) Dissolve 7.5 mg of DCPIP in 50 ml of distilled water. **▲ CRITICAL** Freshly prepare and keep protected from light.

DUB solution in DMSO (12.5 mM) Dissolve 10 mg of DUB in 2.48 ml of DMSO. Store in 500- μl aliquots at -20°C for several months. **▲ CRITICAL** Keep protected from light.

Malonic acid solution (1 M) Dissolve 104 mg of malonic acid in 1 ml of distilled water and store it at 4°C for several weeks.

TTFA solution (50 mM) Dissolve 11.1 mg of TTFA in 1 ml of DMSO and store it at 4°C for several months.

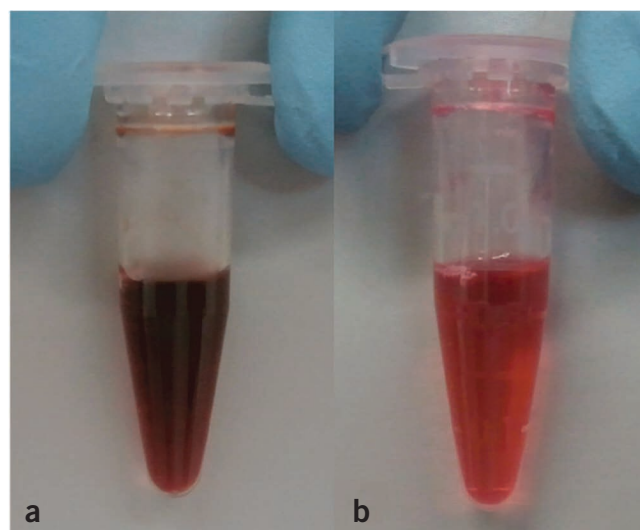


Figure 3 | Cytochrome *c* reduction. **(a)** Oxidized cytochrome *c* solution (brown color). **(b)** Cytochrome *c* solution after reduction with sodium dithionite (orange-pink color).

Tween-20 solution (2.5% (vol/vol)) Dissolve 200 μl of Tween-20 in 7.8 ml of distilled water. **▲ CRITICAL** Freshly prepare and keep protected from light.

Antimycin A stock solution (10 mg ml⁻¹) Dissolve 25 mg of antimycin A in 2.5 ml of ethanol and store at -20°C . Dilute the stock solution to 1 mg ml⁻¹ by adding 10 μl of stock solution to 90 μl of ethanol on the day of the analysis.

! CAUTION Antimycin A is highly toxic; avoid skin contact and inhalation.

DUB solution in ethanol (10 mM) Dissolve 25 mg of DUB in 7.74 ml of absolute ethanol and store in aliquots at -20°C for several months.

Decylubiquinol solution (10 mM) Add a few grains of potassium borohydride to 250 μl of 10 mM DUB in ethanol. Add 5- μl aliquots of 0.1 M HCl until the solution becomes colorless. Briefly spin the solution at 10,000g for 1 min and transfer the solution into a new 500- μl tube, avoiding any potassium borohydride crystals. Adjust the pH of the solution between 2 and 3 with 5- μl aliquots of 1 M HCl and keep the solution on ice protected from light (see TROUBLE-SHOOTING). **▲ CRITICAL** This solution should be freshly prepared.

! CAUTION Potassium borohydride is highly toxic; avoid skin contact and inhalation.

EDTA solution (5 mM) Dissolve 46.5 mg of EDTA in 20 ml of distilled water. Adjust the pH to 7.5 with sodium hydroxide and the volume to 25 ml. Store at room temperature (20 – 25°C) for several weeks.

DTNB solution (1 mM) Dissolve 7.9 mg of DTNB in 20 ml of 100 mM Tris (pH 8.0). **▲ CRITICAL** This solution should be freshly prepared.

Ac CoA solution (10 mM) Dissolve 100 mg Ac CoA in 12.35 ml of distilled water. Store at -80°C in 200- μl aliquots for several months.

▲ CRITICAL Once thawed, Ac CoA aliquots should be used on the same day.

Oxaloacetic acid solution (10 mM) Dissolve 6.6 mg of oxaloacetic acid in 5 ml of distilled water. **▲ CRITICAL** The solution should be freshly prepared.

PROCEDURE

Sample preparation for RC enzyme assays

1 | We describe three possible sample preparation methods (options A, B and C):

- Option A Preparation of a skeletal muscle homogenate
- Option B Preparation of fibroblast lysate
- Option C Preparation of mitochondrial-enriched fractions from fibroblasts

(A) Skeletal muscle homogenization ● TIMING ~30 min

- (i) Weigh about 30–50 mg of frozen skeletal muscle (stored at -80°C or in liquid nitrogen), remove any visible fat and connective tissue with a scalpel blade and dissect it in fragments as small as possible (possibly having a diameter

smaller than 0.5 mm). The use of a tissue chopper, if available, will be helpful³². Dilute 1:20 in ice-cold sucrose muscle homogenization buffer in a 1-ml glass-glass tissue grinder.

▲ **CRITICAL STEP** Precool the tissue grinder in ice for 5 min before starting the homogenization. Homogenization, as well as all the following steps, should be done on ice to avoid loss of activity as a result of proteases.

- (ii) Homogenize the muscle using a clean glass-glass conical tissue grinder kept on ice with 15 slow and controlled up-down strokes at 500 r.p.m. (**Supplementary Video 1**).

▲ **CRITICAL STEP** A good compliance between pestle and vial is imperative to achieve a reproducible homogenization and to avoid enzymatic inactivation from heat generation and excessive shearing forces.

▲ **CRITICAL STEP** Tissue grinders must be carefully cleaned and dried before the analysis in order to avoid chemical and microbial contamination. After each homogenization, perform appropriate disinfection: thoroughly clean the tissue grinder with a brush using a detergent (e.g., Alconox), rinse several times with hot water; finally, rinse with distilled water.

- (iii) Centrifuge the muscle homogenate at 600g for 10 min at 4 °C.

- (iv) Transfer the supernatant into a new tube on ice for the RC analysis. Keep a 10-μl aliquot for total protein concentration measurements according to the Bradford method³³.

▲ **CRITICAL STEP** This method normally yields a total protein concentration of 2–3 mg ml⁻¹ in the muscle supernatant.

■ **PAUSE POINT** The muscle supernatant is now ready to be used; keep the preparation on ice and use it on the same day. Flash-freezing of the supernatant in liquid nitrogen and storage at –80 °C will result in a limited loss of enzymatic activity (usually <15%).

(B) Preparation of fibroblast lysate ● TIMING ~30 min

- (i) Remove the medium from about 5 × 10⁶ cells and wash them once with PBS. Keep the removed medium in a separate tube.
- (ii) Detach the cells using 0.05% (wt/vol) trypsin-EDTA, and after cell detachment block the trypsin by adding the removed medium. Transfer the cell suspension to a 15-ml conical tube. Count cells using a counting chamber and verify that the number of cells is not less than 5 × 10⁶ cells.
- (iii) Wash the cells by centrifuging at 1,000g for 5 min at 4 °C, discarding the supernatant and resuspending the cells in 3 ml of PBS.
- (iv) Wash the cells as described in Step 1B(iii).
- (v) Centrifuge the cells at 1,000g for 5 min at 4 °C and discard the supernatant.
- **PAUSE POINT** The fibroblast pellet can be flash-frozen in liquid nitrogen and stored at –80 °C for weeks.
- (vi) Suspend the fibroblast pellet in 0.4 ml of 20 mM hypotonic potassium phosphate buffer (pH 7.5).
- (vii) By using a 50-μl Hamilton syringe, take up and expel the suspension several times until it has the appearance of a homogeneous solution.
- (viii) Snap-freeze the cell lysate in liquid nitrogen and thaw it at 37 °C three times and keep it on ice for the RC analysis. Keep a 10-μl aliquot for total protein concentration measurements according to the Bradford method.

▲ **CRITICAL STEP** This method commonly yields a total protein concentration of ~2 mg ml⁻¹ in the fibroblast solution.

■ **PAUSE POINT** Fibroblast solution is now ready to be used; keep the preparation on ice and use on the same day.

(C) Preparation of mitochondrial-enriched fractions from fibroblasts ● TIMING ~1 h 30 min

- (i) Remove the medium from about 10–15 × 10⁶ human fibroblasts and wash them once with PBS. Keep the removed medium in a separate tube.
- (ii) Detach the cells using 0.05% (wt/vol) trypsin-EDTA, and after cell detachment, block the trypsin by adding the removed medium. Transfer the cell suspension in a 15-ml polypropylene tube. Count the cells using a counting chamber.
- (iii) Wash the cells by centrifuging at 1,000g for 5 min at 4 °C, discarding the supernatant and resuspending the cells in 3 ml of PBS.
- (iv) Wash the cells as described in Step 1C(iii).
- (v) Centrifuge the cells at 1,000g for 5 min at 4 °C.
- (vi) Discard the supernatant and flash-freeze the fibroblast pellet in liquid nitrogen.
- (vii) Thaw the fibroblast pellet and suspend it in 1 ml of 10 mM ice-cold hypotonic Tris buffer (pH 7.6).
- (viii) Homogenize with a 2-ml glass/Teflon tissue grinder with a tight clearance kept on ice with 15 slow up-down strokes at 1,800 r.p.m.

▲ **CRITICAL STEP** Precool the tissue grinder on ice for 5 min before starting the homogenization.

▲ **CRITICAL STEP** Tissue grinders must be carefully cleaned and dried before the analysis to avoid chemical and microbial contamination. After each homogenization, perform appropriate disinfection, thoroughly clean the tissue grinder with a brush using a detergent (i.e., Alconox), rinse several times with hot water; finally, rinse with distilled water.

▲ **CRITICAL STEP** A good compliance between pestle and vial is imperative in order to achieve an effective homogenization.

- (ix) Add 200 μl of 1.5 M sucrose solution to the cell homogenate and mix thoroughly.
- (x) Centrifuge at 600g for 10 min at 2 °C.
- (xi) Collect the supernatant, transfer it to a 1.5-ml microcentrifuge tube and centrifuge at 14,000g for 10 min at 2 °C.
- (xii) Carefully discard the supernatant and resuspend the mitochondrial pellet in 0.5 ml of 10 mM ice-cold hypotonic Tris buffer (pH 7.6).
- (xiii) Divide the mitochondrial solution in aliquots, flash-freeze them in liquid nitrogen and store at -80°C . Keep a 10- μl aliquot for total protein concentration measurements according to the Bradford method.
- **PAUSE POINT** Frozen mitochondria can be stored at -80°C for several days.
- (xiv) Thaw the cell mitochondrial solution and subject it to three cycles of freeze-thawing in liquid nitrogen to disrupt the mitochondrial membranes just before use.

Spectrophotometric RC enzyme analysis ● **TIMING** ~2 h 30 min

2| There are seven assay types, which are summarized below (see also **Table 1**):

Option	Complex(es) analyzed	Comment
A	Complex I (CI)	NADH:ubiquinone oxidoreductase
B	Complex II (CII)	Succinate dehydrogenase
C	Complex III (CIII)	Decylubiquinol cytochrome <i>c</i> oxidoreductase
D	Complex IV (CIV)	Cytochrome <i>c</i> oxidase
E	Complex I + III (CI+III)	NADH cytochrome <i>c</i> oxidoreductase: this combined assay can be helpful for an indirect detection of coenzyme Q deficiency. This assay is not reliable in cultured fibroblasts and in liver tissue
F	Complex II + III (CII + III)	Succinate cytochrome <i>c</i> reductase: this combined assay can be helpful for an indirect detection of coenzyme Q deficiency
G	CS	This is a mitochondrial matrix enzyme, which is commonly used to normalize the results of the assays for the respiratory chain enzymes (CI – CIV and CI+III, CII + III)

(A) Complex I (CI, NADH:ubiquinone oxidoreductase)

- (i) Add the sample (10–40 μg of proteins from mouse muscle homogenate, 2–10 μg of mouse muscle mitochondria or 20–50 μg of isolated mitochondria from human fibroblasts (**Fig. 4**)) to 700 μl of distilled water in a 1-ml cuvette.
- ▲ **CRITICAL STEP** Isolated mitochondria should be subjected to three cycles of freeze-thawing in hypotonic buffer before measuring complex I in order to maximize the enzymatic rates (**Fig. 4**).
- (ii) Add 100 μl of potassium phosphate buffer (0.5 M, pH 7.5), 60 μl of fatty acid-free BSA (50 mg ml^{-1}), 30 μl of KCN (10 mM) and 10 μl of NADH (10 mM).
- (iii) Adjust the volume to 994 μl with distilled water.
- (iv) Prepare, in parallel, a separate cuvette containing the same quantity of reagents and sample but with the addition of 10 μl of 1 mM rotenone solution.
- ! **CAUTION** Rotenone and KCN are toxic; see REAGENT SETUP for details.
- ▲ **CRITICAL STEP** In order to avoid underestimation of rotenone-resistant activities, incubation of rotenone in a separate cuvette in parallel is crucial.
- (v) Mix by inverting the cuvette covered with Parafilm and read the baseline at 340 nm for 2 min.
- (vi) Start the reaction by adding 6 μl of ubiquinone₁ (10 mM), mix by inverting the cuvette using Parafilm and follow the decrease of absorbance at 340 nm for 2 min. Specific complex I activity is the rotenone-sensitive activity. A representative trace is illustrated in **Figure 5a**.

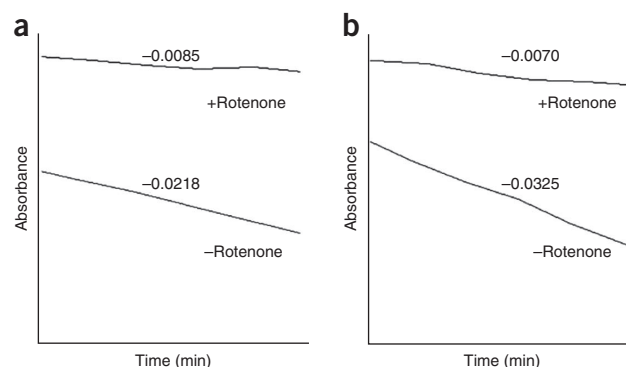


Figure 4 | Effect of freeze-thaw cycles on complex I activities in fibroblast-isolated mitochondria. (a,b) Thirty micrograms of mitochondria from human fibroblasts were assayed with and without rotenone in parallel cuvettes, without (a) or with (b) three freeze-thaw cycles in hypotonic buffer (Tris 10 mM, pH 7.6). Repeated freeze-thaw cycles increased the rotenone-sensitive activity by about 90%.

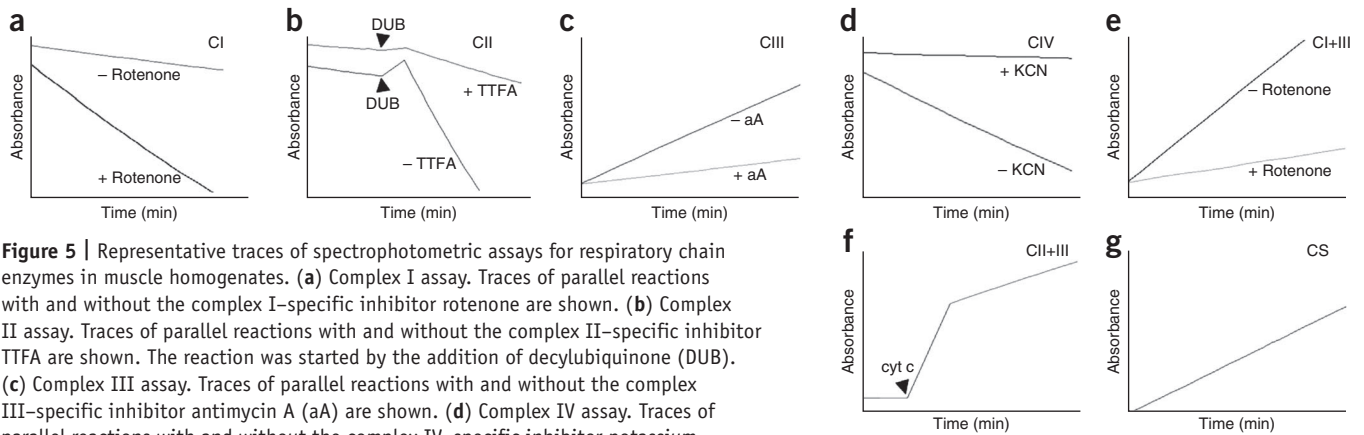


Figure 5 | Representative traces of spectrophotometric assays for respiratory chain enzymes in muscle homogenates. **(a)** Complex I assay. Traces of parallel reactions with and without the complex I-specific inhibitor rotenone are shown. **(b)** Complex II assay. Traces of parallel reactions with and without the complex II-specific inhibitor TTFA are shown. The reaction was started by the addition of decylubiquinone (DUB). **(c)** Complex III assay. Traces of parallel reactions with and without the complex III-specific inhibitor antimycin A (aA) are shown. **(d)** Complex IV assay. Traces of parallel reactions with and without the complex IV-specific inhibitor potassium cyanide (KCN) are shown. **(e)** Complex I + III coupled assay. Traces of parallel reactions with and without the complex I-specific inhibitor rotenone are shown. **(f)** Complex II + III assay. The reaction is started by the addition of cytochrome *c* (cyt *c*). **(g)** Citrate synthase (CS) assay.

▲ **CRITICAL STEP** Discard the Parafilm after each mixing step and avoid contamination of different cuvettes with rotenone.

? TROUBLESHOOTING

(B) Complex II (CII, succinate dehydrogenase)

- To a 1-ml cuvette, add 600 μ l of distilled water, 50 μ l of potassium phosphate buffer (0.5 M, pH 7.5), 20 μ l of fatty acid-free BSA (50 mg ml^{-1}), 30 μ l of KCN (10 mM), 50 μ l of succinate (400 mM), the sample (2–10 μ g of mouse muscle homogenate, 0.5–2 μ g of mouse muscle mitochondria or 8–60 μ g of total human fibroblast lysate) and 145 μ l of DCPIP (0.015% (wt/vol)); adjust the volume to 996 μ l with distilled water.

! **CAUTION** KCN is highly toxic; see REAGENTS for details.

- Mix by inverting the cuvette and incubate inside the spectrophotometer at 37 °C for 10 min. Read the baseline activity at 600 nm for the last 2 min.

▲ **CRITICAL STEP** Preincubation of the sample with succinate is mandatory to fully activate the enzyme.

- Start the reaction by adding 4 μ l of 12.5 mM DUB, mix by inversion and follow the decrease of absorbance at 600 nm for 3 min.

- Check the specificity of complex II activity by running the assay after the addition of 10 μ l of either 1 M malonate or 50 mM TTFA before starting the reaction. The degree of inhibition usually exceeds 85% with TTFA and 95% with malonate, indicating a high specificity of the reaction. A representative trace is illustrated in **Figure 5b**.

? TROUBLESHOOTING

(C) Complex III (CIII, decylubiquinol cytochrome *c* oxidoreductase)

- To a 1-ml cuvette, add 730 μ l of distilled water, 50 μ l of potassium phosphate buffer (0.5 M, pH 7.5), 75 μ l of oxidized cytochrome *c*, 50 μ l of KCN (10 mM), 20 μ l of EDTA (5mM, pH 7.5), 10 μ l of Tween-20 (2.5% (vol/vol)) and the sample (0.6–3 μ g of mouse muscle supernatant proteins, 0.1–1 μ g of mouse muscle mitochondria or 2–20 μ g of total human fibroblast lysate proteins).

▲ **CRITICAL STEP** Addition of EDTA dramatically increases the sensitivity of the assay.

- Prepare, in parallel, a separate cuvette containing the same quantity of reagents and sample with the addition of 10 μ l of 1 mg ml^{-1} antimycin A solution.

! **CAUTION** Antimycin A and KCN are highly toxic; see REAGENT SETUP for details.

▲ **CRITICAL STEP** Incubation of antimycin A in a separate cuvette in parallel is crucial in order to avoid underestimation of antimycin A-resistant activities.

- Adjust the volume to 990 μ l with distilled water.

- Mix by inverting the cuvettes and read the baseline at 550 nm for 2 min.

- Start the reaction by adding 10 μ l of 10 mM decylubiquinol, mix rapidly by inverting the cuvette using Parafilm, and then immediately observe the increase in absorbance at 550 nm for 1–2 min. Specific complex III activity is the antimycin A-sensitive activity. A representative trace is illustrated in **Figure 5c**.

▲ **CRITICAL STEP** Discard the Parafilm after each mixing step and avoid contamination between different cuvettes with antimycin A.

▲ **CRITICAL STEP** The linearity of the reaction is limited to 1–2 min: no more than two or three cuvettes may be analyzed at the same time.

? TROUBLESHOOTING

(D) Complex IV (CIV, cytochrome *c* oxidase)

- (i) To a 1-ml cuvette, add 400 μ l of distilled water, potassium phosphate buffer (100 mM, pH 7.0; 500 μ l for muscle, 250 μ l for cells), reduced cytochrome *c* (1 mM; 60 μ l for muscle, 50 μ l for cells) and read the baseline activity at 550 nm for the last 2 min.
- (ii) Adjust the volume to 995 μ l with distilled water.
- (iii) Start the reaction by adding 5 μ l of sample (0.4–2.5 μ g of mouse muscle supernatant proteins, 0.1–1 μ g of mouse muscle mitochondria or 5–40 of human fibroblast lysate), mix by inversion and monitor the decrease of absorbance at 550 nm. This takes 3 min.
▲ CRITICAL STEP Dilution of the sample in homogenization buffer may be necessary in order to increase the precision of pipetting the limited amount of sample required for the analysis.
- (iv) To check the specificity of complex IV activity, add 30 μ l of 10 mM KCN in a separate reaction prepared as described Step 2D(i–iii). However, specificity is close to maximal both in tissue and cells. A representative trace is illustrated in **Figure 5d**.

! CAUTION KCN is highly toxic; see REAGENT SETUP for details.

? TROUBLESHOOTING

(E) Complex I + III (CI + III, NADH cytochrome *c* oxidoreductase)

- (i) Incubate the muscle supernatant (4–25 μ g of mouse muscle homogenates or 1–6 μ g of mouse muscle mitochondria) in 700 μ l of distilled water in a 1-ml cuvette for 2 min to induce an osmotic shock³⁴.
- (ii) Add 100 μ l of potassium phosphate buffer (0.5 M, pH 7.5), 20 μ l of fatty acid-free BSA (50 mg ml⁻¹), 30 μ l of KCN (10 mM) and 50 μ l of oxidized cytochrome *c* (1 mM). Adjust the volume to 980 μ l with distilled water.
! CAUTION Rotenone and KCN are toxic; see REAGENT SETUP for details.
- (iii) Prepare, in parallel, a separate cuvette containing the same quantity of reagents and sample plus 10 μ l of 1 mM rotenone solution.
▲ CRITICAL STEP Incubation of rotenone in a separate cuvette in parallel is crucial in order to avoid underestimation of rotenone-resistant activities.
- (iv) Mix by inversion and read the baseline at 550 nm for 2 min.
- (v) Start the reaction by adding 20 μ l of 10 mM NADH, mix by inversion using Parafilm, and then follow the increase of absorbance at 550 nm for 2 min. Specific complex I+III activity is the rotenone-sensitive activity. A representative trace is illustrated in **Figure 5e**.

▲ CRITICAL STEP Discard the Parafilm after each mixing step and avoid contamination of different cuvettes with rotenone.

? TROUBLESHOOTING

(F) Complex II + III (CII + III, succinate cytochrome *c* reductase)

- (i) To a 1-ml cuvette, add 800 μ l of distilled water, potassium phosphate buffer (0.5 M, pH 7.5; 40 μ l for muscle, 100 μ l for cells), 30 μ l of KCN (10 mM), 25 μ l of succinate (400 mM) and the sample (1.5–10 μ g of mouse muscle supernatant proteins, 0.2–1 μ g of mouse muscle mitochondria or 15–80 μ g of human fibroblast lysate); adjust the volume to 950 μ l with distilled water.

! CAUTION KCN is highly toxic; see REAGENT SETUP for details.

- (ii) Mix by inverting the cuvette and incubate it for 10 min inside the spectrophotometer at 37 °C.
▲ CRITICAL STEP Preincubation of the sample with succinate for 10 min is mandatory in order to fully activate the enzyme
- (iii) Start the reaction by adding 50 μ l of oxidized cytochrome *c* (1 mM), mix by inversion, and then follow the increase of absorbance at 550 nm for 3 min.
- (iv) Check the specificity of this assay by adding 10 μ l of 1 M malonate or 1 mg ml⁻¹ antimycin A in a separate cuvette prepared as described Step 2F(i–iii), but this is close to maximal¹⁷. A representative trace is illustrated in **Figure 5f**.

? TROUBLESHOOTING

(G) Citrate synthase

- (i) To a 1-ml cuvette, add 300 μ l of distilled water, 500 μ l of Tris (200 mM, pH 8.0) with Triton X-100 (0.2% (vol/vol)), 100 μ l of DTNB, 30 μ l of Ac CoA (10 mM) and the sample (1–6 μ g of mouse muscle supernatant proteins, 0.2–1.5 μ g of mouse muscle mitochondria or 5–40 μ g of human fibroblast lysate); adjust the volume to 950 μ l with distilled water.
- (ii) Mix by inversion and read the baseline activity at 412 nm for 2 min.
- (iii) Start the reaction by adding 50 μ l of 10 mM oxaloacetic acid, mix by inversion, and then monitor the increase in absorbance at 412 nm for 3 min. A representative trace is illustrated in **Figure 5g**.

? TROUBLESHOOTING

? TROUBLESHOOTING

Troubleshooting advice can be found in **Table 2**. See also **Figure 6**. All of the problems listed in the table will become evident at the end of the assays described in Step 2.

TABLE 2 | Troubleshooting table.

Step	Problem	Possible reason	Solution
2	Overall reduction of RC enzymatic activities in muscle or cells	Deterioration of muscle tissue or cultured cells	Assure appropriate sampling, handling and storage of the muscle after flash-freezing in nitrogen. Check the appearance and degree of confluence of the cultured cells with a microscope before collecting. Always include one or more control samples with known RC activities in the experiments
		Contamination from inappropriately cleaned glassware	Carefully clean the tissue grinders with a glassware detergent (i.e., Alconox) using a brush; rinse several times with hot water and finally with distilled water
		Insufficient, excessive or inappropriate homogenization	The up-down strokes should be performed smoothly and consistently to avoid unpredictable shearing forces. The compliance between pestle and mortar should be carefully verified before use (Fig. 1b–e)
		The temperature controller might be malfunctioning	Verify the temperature using a cuvette filled with water
		Malfunctioning spectrophotometer lamp	Verify the function of the lamp or replace it
	Nonlinear reactions	Excessive amount of sample used	Reduce the quantities of sample for the assay until linearity of the reaction is observed for at least 1 min (Fig. 6)
	No reaction is observed	Insufficient amount of sample used	Increase the amount of sample until a clear reaction is observed
	Irregular shape of the plots	The tissue/cell lysate is not homogeneous	Verify that the lysis step has been performed correctly and that no cell debris is appreciable
	2A,2E	Low complex I+II activities	NADH may be oxidized
		Rotenone contamination has occurred between cuvettes during mixing by inversion	Verify that NADH has not been oxidized: the color should be light whitish yellow Discard Parafilm each time after inverting a cuvette with rotenone
2A	Low rotenone sensitivity	Cultured cell lysate has been used	Complex I activity cannot be measured on total cell lysate; use isolated mitochondria
		Isolated mitochondria has not been freeze-thawed	Perform three rounds of freeze-thawing in hypotonic buffer as described in Step 1C(xv)
2B,2F	Low complex II and II+III activities	The sample has not been incubated with succinate	Incubate the sample with succinate as described
2C	Low/absent complex III activities	Decylubiquinone has not been efficiently reduced	Verify that decylubiquinol is colorless; store it on ice and protect it from light
		Antimycin A contamination has occurred between cuvettes during inversion	Discard the Parafilm each time after inverting a cuvette with antimycin A
		Spectrophotometric recording has been started with excessive delay after starting the reaction and reaction rates have decayed within minutes	Assay up to 2–3 cuvettes at the same time; start recording immediately after adding decylubiquinol and inverting
2D	Low/absent complex IV activities	Cytochrome <i>c</i> has not been efficiently reduced	Efficient reduction of cytochrome <i>c</i> should be verified as indicated in REAGENT SETUP and in Box 1
2G	Low citrate synthase activities	Acetyl CoA, oxaloacetic acid or DTNB has deteriorated	Verify that storage conditions of acetyl CoA and oxaloacetic acid are appropriate. DTNB and oxaloacetic solutions should be freshly prepared

TIMING

Step 1: ~30 min for options A and B, and ~1 h 30 min for option C (Muscle should be previously collected and usually stored at -80°C ; cells will have to be seeded a few days before the analysis until the required number of cells is reached.)

Step 2: ~2 h 30 min (Note that several solutions have to be prepared on the day of the analysis, as indicated; the timing stated is for one sample in duplicate.)

ANTICIPATED RESULTS

These protocols will allow a reliable, sensitive and specific estimate of the catalytic activities of the RC enzymes in tissues and cells. The relative standard deviation (the ratio of the standard deviation to the mean expressed as a percentage) of repeated measurements of the same tissue homogenate will be within 7%, whereas the relative standard deviation of the RC results obtained by the same muscle homogenized at different times will be within 15% (ref. 17). The specificity of the enzymatic rates is controlled by the use of the specific enzyme inhibitors for each complex.

The enzymatic activities for each mitochondrial enzyme should be calculated as $\text{nmol min}^{-1} \text{mg}^{-1}$ of protein according to the following equation (refer to **Table 1** for the summary of the substrates used in the different assays):

enzyme activity ($\text{nmol min}^{-1} \text{mg}^{-1}$) = $(\Delta \text{Absorbance}/\text{min} \times 1,000)/[(\text{extinction coefficient} \times \text{volume of sample used in ml}) \times (\text{sample protein concentration in } \text{mg ml}^{-1})]$.

The specific activity of complex I is calculated by subtracting total complex I activity (without rotenone) and rotenone-resistant activity (with rotenone), and it is expressed as $\text{nmol min}^{-1} \text{mg}^{-1}$ of total proteins (extinction coefficient for NADH $6.2 \text{ mM}^{-1} \text{ cm}^{-1}$). Specificity of complex I activity, estimated by the percentage of inhibition by rotenone, will be >80% in muscle and >65% in isolated mitochondria from cultured cells.

Complex II activity is expressed as $\text{nmol min}^{-1} \text{mg}^{-1}$ of total proteins (extinction coefficient for DCPIP $19.1 \text{ mM}^{-1} \text{ cm}^{-1}$). Specificity of complex II activity, estimated by the percentage of inhibition by the addition of specific complex II inhibitors, will be over 87% when using TTFA and over 98% when using malonate, in either muscle or total cell lysates.

The specific activity of complex III is calculated by subtracting total complex III activity (without antimycin A) and antimycin A-resistant activity (with antimycin A), and it is expressed as $\text{nmol min}^{-1} \text{mg}^{-1}$ of total proteins (extinction coefficient for reduced cytochrome *c* $18.5 \text{ mM}^{-1} \text{ cm}^{-1}$). Specificity of complex III activity, estimated by the percentage of inhibition by antimycin A, will be often over 60% in muscle and in total cell lysates, depending on the amount of sample proteins.

Complex IV activity is expressed as $\text{nmol min}^{-1} \text{mg}^{-1}$ of total proteins (extinction coefficient for reduced cytochrome *c* $18.5 \text{ mM}^{-1} \text{ cm}^{-1}$). The specificity of complex IV activity, estimated by the percentage of inhibition with KCN, will be over 95% in muscle and in total cell lysates.

The specific activity of complex I+III is calculated by subtracting total complex I activity (without rotenone) and rotenone-resistant activity (with rotenone), and it is expressed as $\text{nmol min}^{-1} \text{mg}^{-1}$ of total proteins (extinction coefficient for reduced cytochrome *c* $18.5 \text{ mM}^{-1} \text{ cm}^{-1}$).

The specificity of complex I+III activity, estimated by the percentage of inhibition by rotenone, will be usually over 50% in muscle, depending on the muscle protein concentration. Complex II+III activity is expressed as $\text{nmol min}^{-1} \text{mg}^{-1}$ of total proteins (extinction coefficient for reduced cytochrome *c* $18.5 \text{ mM}^{-1} \text{ cm}^{-1}$). CS activity is expressed as $\text{nmol min}^{-1} \text{mg}^{-1}$ of total proteins (extinction coefficient $13.6 \text{ mM}^{-1} \text{ cm}^{-1}$).

The enzymatic activities of the RC complexes are also commonly further normalized to the activity of CS, a mitochondrial matrix enzyme, used as a marker of the abundance of mitochondria within a tissue/cell. This strategy is particularly useful to detect partial respiratory enzymatic dysfunction associated with compensatory mitochondrial proliferation, which might be overlooked by considering only the enzymatic activities normalized to protein content.

Supplementary Table 1 shows example results from lysates and isolated mitochondria of mouse skeletal muscle and human fibroblasts. Some examples of distinct RC enzymatic defects in cells from patients with different mitochondrial disorders are illustrated in **Supplementary Figure 1**.

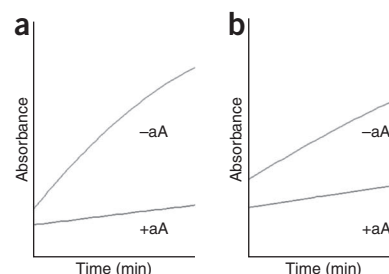


Figure 6 | Troubleshooting: decreased linearity of the reaction with time because of excessive sample protein concentrations. (a,b) Complex III assay representative traces followed for 2 min obtained from 20 (a) and 5 μg (b) of total muscle protein from muscle homogenates. The limited linearity of trace A was remarkably improved by decreasing the sample protein concentrations.

Note: Supplementary information is available in the online version of the paper.

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